Analysis of Boston Marathon Qualification Trends 2022–2024

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Introduction

The Boston Marathon is one of the most prestigious road races in the world. Unlike most marathons, entry is not first-come-first-served, runners must meet strict qualifying standards set by the Boston Athletic Association (BAA), which vary by age and gender. Even meeting the standard is not always enough. In recent years, demand has been so high that runners needed to beat their standard by several additional minutes to secure a spot.

This project analyzes over 1 million finisher results from qualifying races held between 2022 and 2024 to answer three questions: How have qualification rates changed over time? Which demographics and race characteristics are associated with qualifying? And how close are non-qualifiers to the cutoff?

Data

Three datasets were obtained from Kaggle: race-level results, race metadata, and BAA qualification standards by age and gender bracket. After filtering the 762 race events in the source data to those flagged as valid qualifying events and joining on age bracket and gender, the final dataset contains 1,071,822 runners across 530 races, with an overall qualification rate of 13.6%.

Trends Over Time

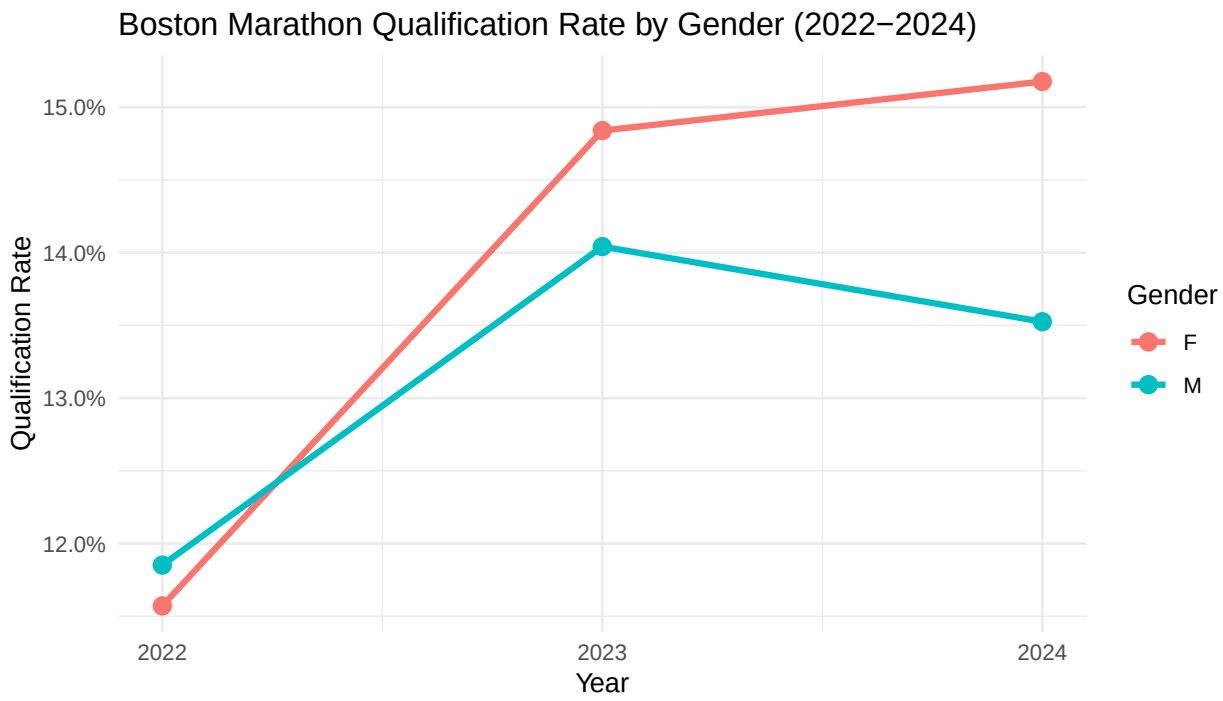
We begin by examining how qualification rates and finish times have evolved across the three-year period. The dataset spans races from the 2022, 2023, and 2024 qualifying cycles, with a notable spike in participation in 2023 due to an increased number of races.

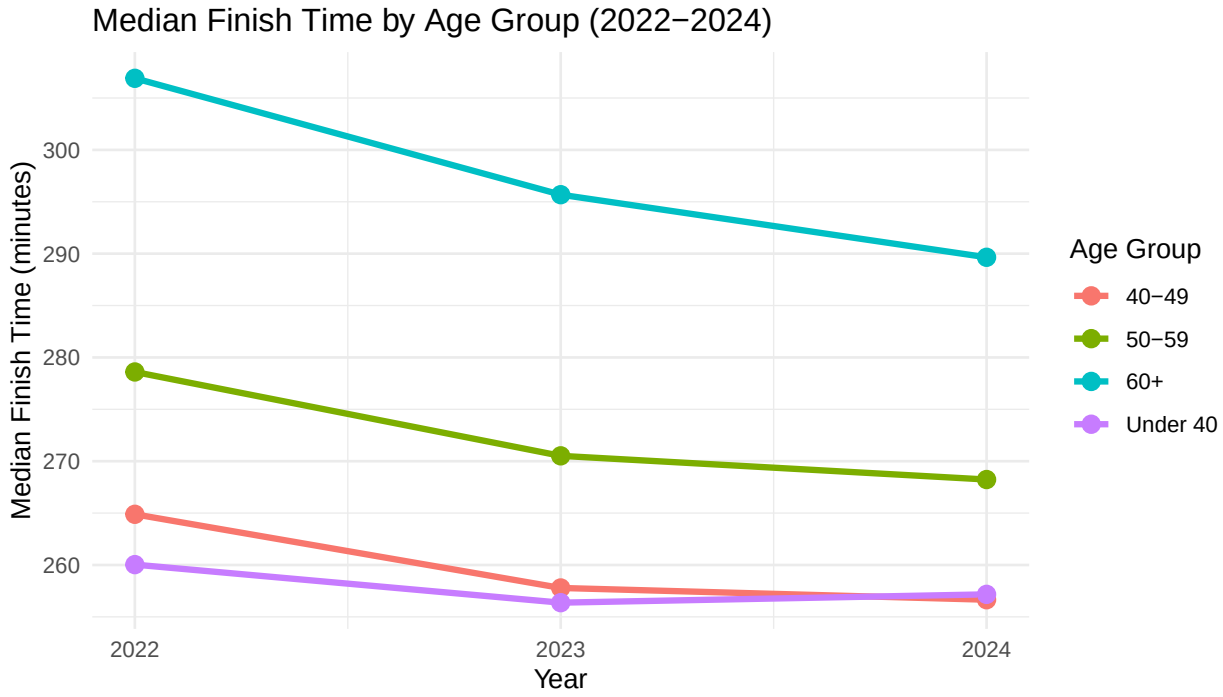
Table 1: Yearly summary of qualification rates and finish times

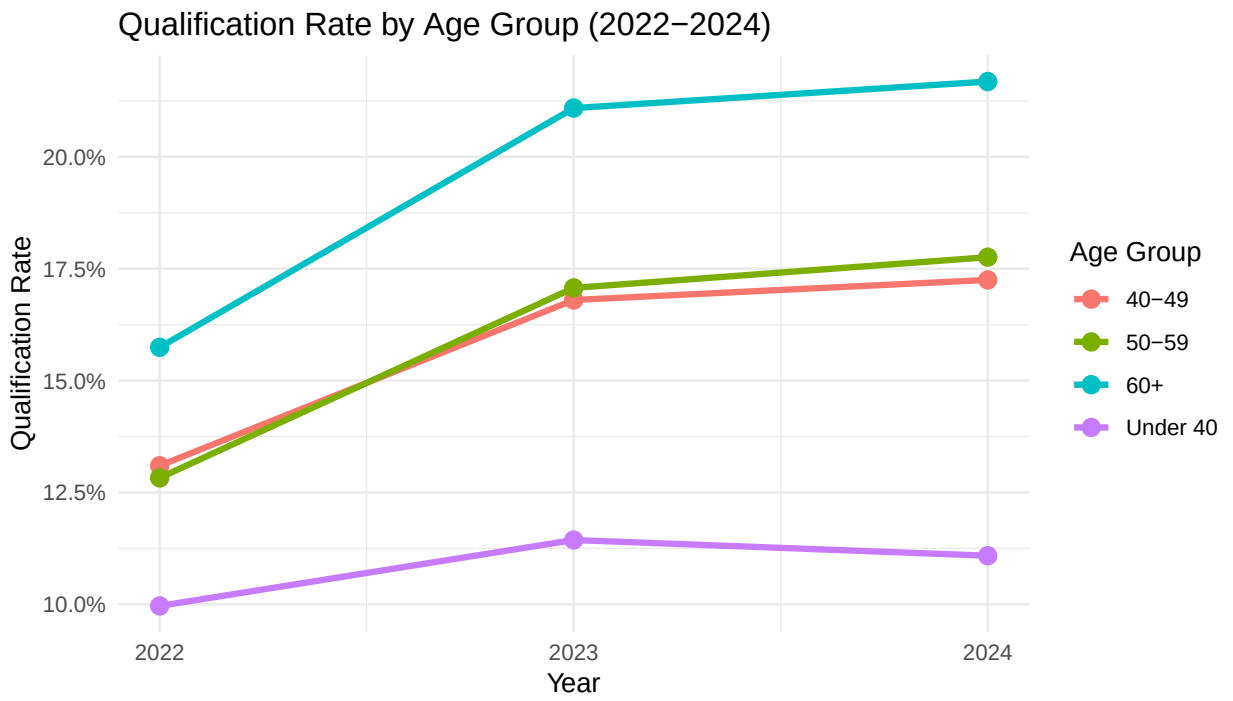
Year	Total Runners	Qualification Rate	Median Finish (min)	Mean Finish (min)	Total Races
2022	286821	11.7%	267.4	276.0	111
2023	517737	14.4%	261.3	270.7	260
2024	267264	14.2%	260.8	270.4	159

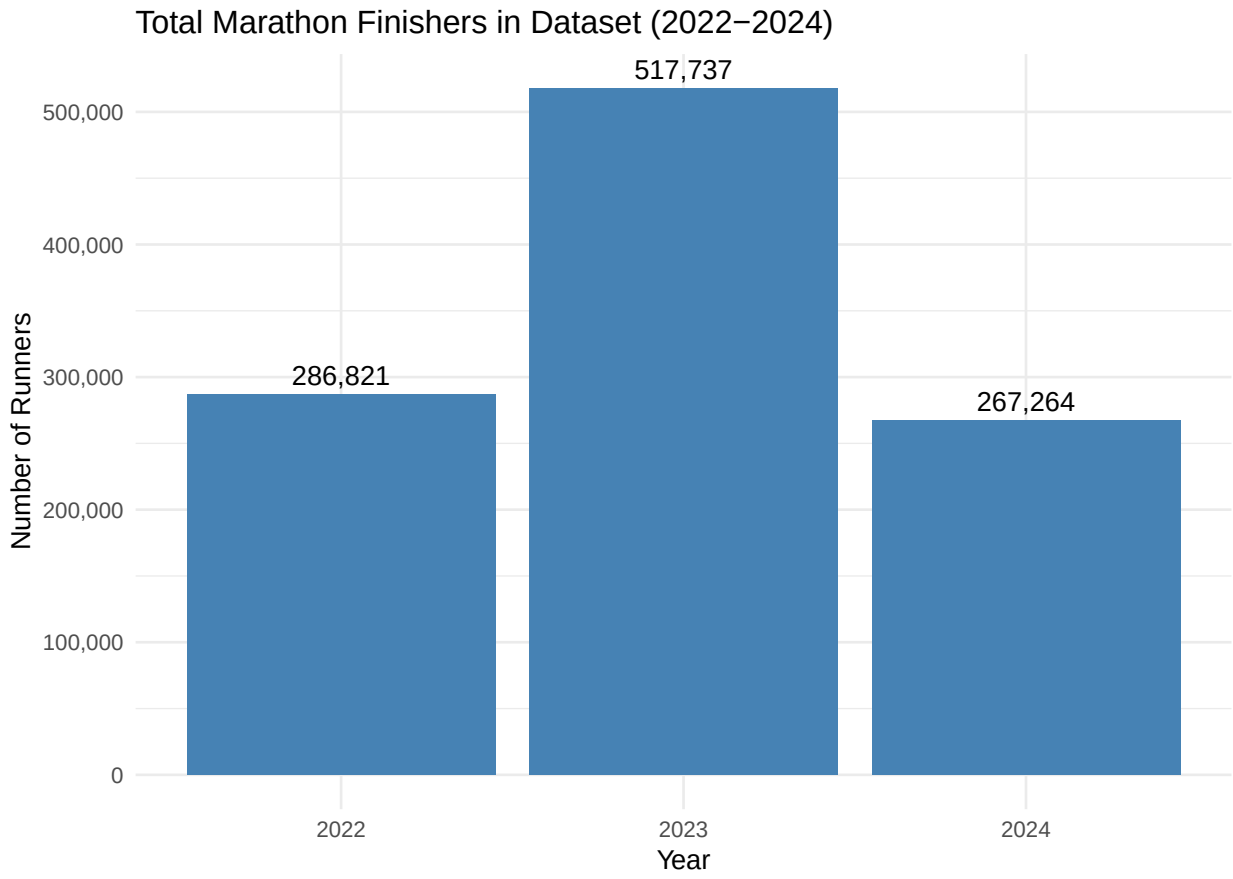
Qualification Rate and Finish Time by Year and Group

Qualification rates rose from 11.7% in 2022 to 14.2% in 2024, accompanied by a 7-minute drop in median finish times. The following charts break these trends down by gender and age group.


Boston Marathon Qualification Rate by Gender (2022−2024)
Qualification Rate
15.0%
14.0%
13.0%
12.0%
2022
2023
2024
Year
Gender
F
M

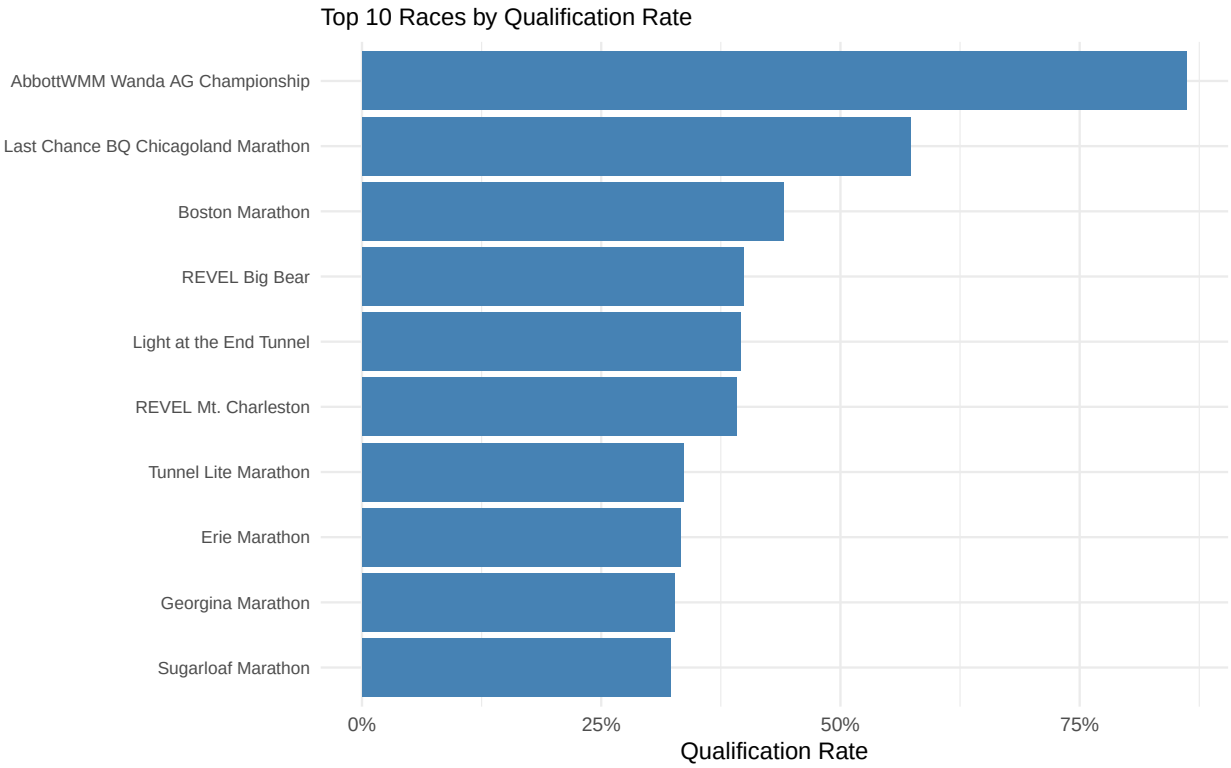

Median Finish Time by Age Group (2022−2024)
Median Finish Time (minutes)
300
290
280
270
260
2022
2023
2024
Year
Age Group
40−49
50−59
60+
Under 40


Qualification Rate by Age Group (2022−2024)
Qualification Rate
20.0%
17.5%
15.0%
12.5%
10.0%
2022
2023
2024
Year
Age Group
40−49
50−59
60+
Under 40


Total Marathon Finishers in Dataset (2022−2024)
Number of Runners
500,000
400,000
300,000
200,000
100,000
0
286,821
517,737
267,264
2022
2023
2024
Year

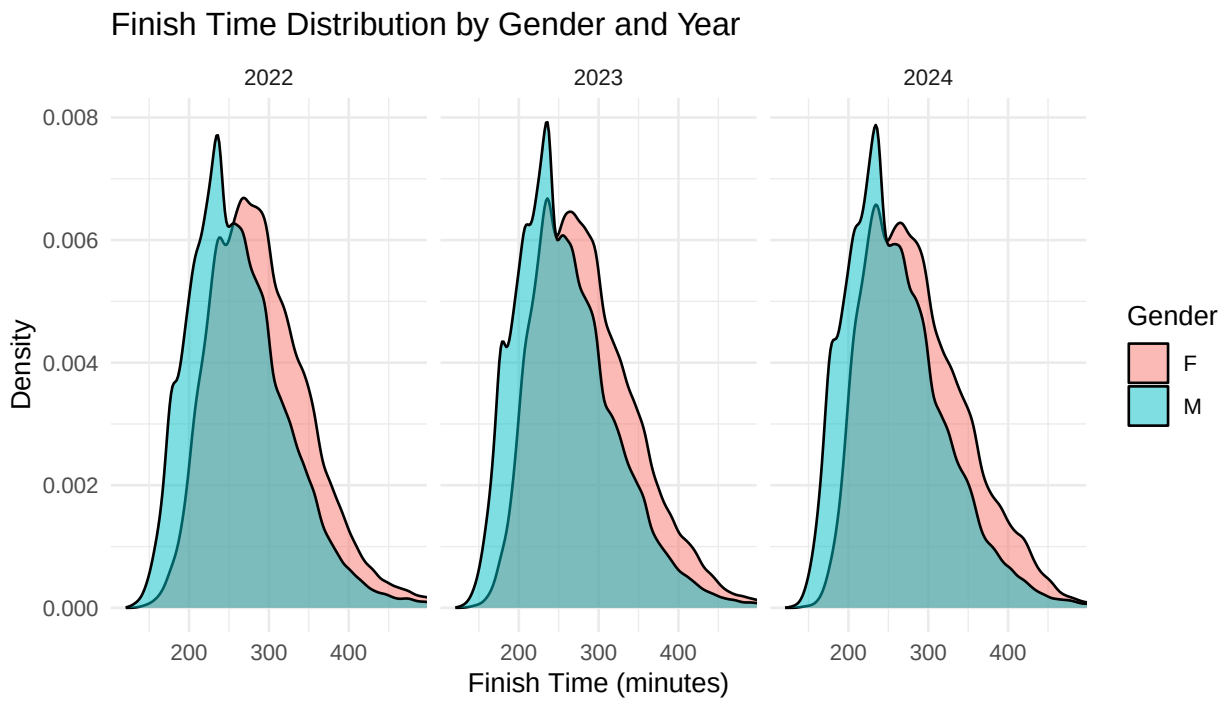
Race-Level Analysis

Not all qualifying races are equal. Certain courses are known for being flat and fast, and some might be more competitive than others. The following analysis examines which races produce the highest qualification rates among races with at least 500 finishes.



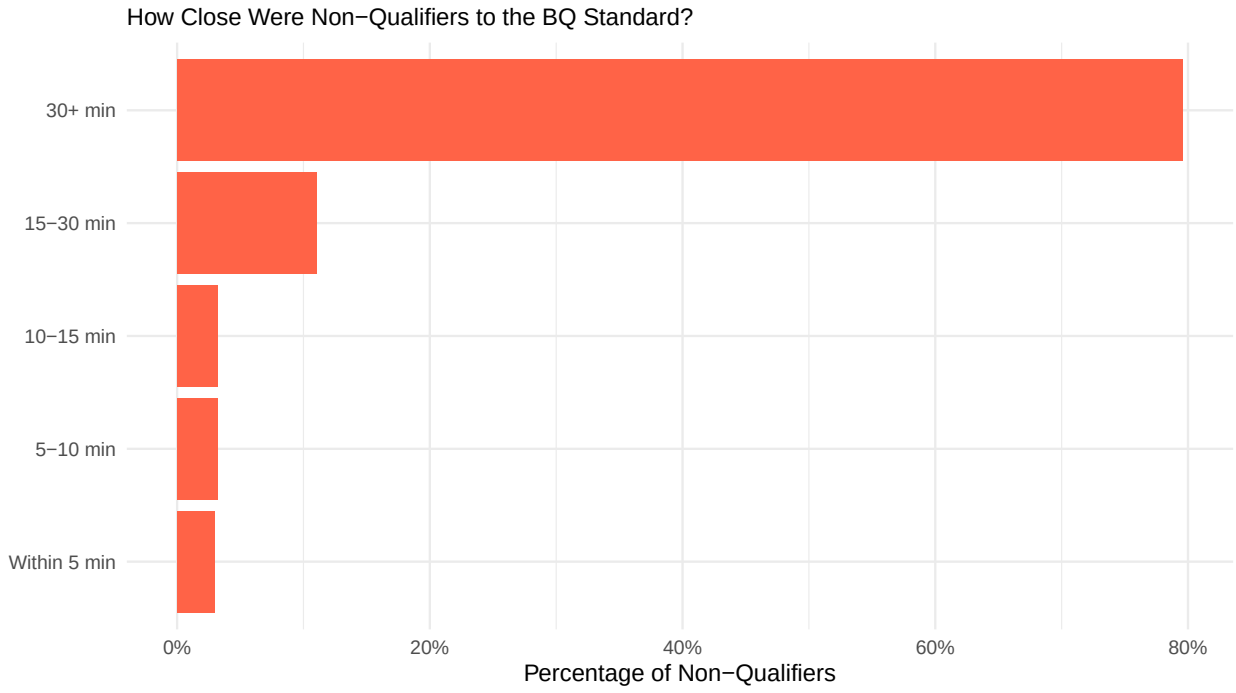
Finish Time Distributions

The density plots below show how finish time distributions have shifted across years, broken down by gender. A leftward shift indicates faster finishing times overall, which is consistent with the trend observed in the yearly summary.



How Close Were Non-Qualifiers?

One of the most interesting questions in this dataset is how close non-qualifiers came to meeting their standard. If most missed by 30+ minutes, the standard is a hard ceiling. If most missed by under 5 minutes, it suggests a large pool of runners on the edge might qualify with modest improvement.



Logistic Regression Models

To quantify the relationship between runner demographics and qualification probability, we fit a series of logistic regression models. We model qualification as a binary outcome (met BQ standard: yes/no) using age, gender, and year as predictors.

Note: finish time was intentionally excluded from the model. Since qualification is deterministically defined as finish time relative to the BQ standard, including it would create target leakage and the model would simply rediscover the cutoff formula rather than learning anything meaningful. This is a known limitation and the models below should be interpreted as capturing demographic and temporal patterns in qualification rather than providing strong individual-level predictions.

Due to the dataset size of over 1 million rows, models were trained on a stratified random sample of 200,000 runners and evaluated on a held-out test set.

Model Comparison

We fit three progressively richer models:

- **Model 1**: Age + Gender
- **Model 2**: Age + Age^2 + Gender (allows non-linear age effect)
- **Model 3**: Age + Age^2 + Gender + Year (adds temporal trend)

Table 2: Model 1 coefficients

Term	Odds Ratio	p-value
(Intercept)	0.161	<0.001
Age_centered	1.014	<0.001
GenderM	0.948	<0.001

Table 3: Model 1 vs Model 2: Chi-square test

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
159994	126787.2	NA	NA	NA
159993	126722.6	1	64.534	<0.001

Table 4: Model 2 vs Model 3: Chi-square test

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
159993	126722.6	NA	NA	NA
159991	126520.6	2	202.065	<0.001

Table 5: AIC comparison across models

	df	AIC
model1	3	126793.2

	df	AIC
model2	4	126730.6
model3	6	126532.6

Coefficients and Odds Ratios

The table below shows the odds ratios from Model 3. An odds ratio above 1 indicates the predictor is associated with a higher probability of qualifying.

Table 6: Model 3 odds ratios

Term	Odds Ratio	95% CI Low	95% CI High	p-value
Year2023	1.274	1.229	1.320	<0.001
Year2024	1.272	1.221	1.325	<0.001
Age_centered	1.013	1.012	1.014	<0.001
I(Age_centeredˆ2)	1.000	1.000	1.000	<0.001
GenderM	0.942	0.915	0.970	<0.001
(Intercept)	0.129	0.124	0.134	<0.001

Mixed Effects Model

To account for the fact that qualification rates vary across races, we fit a mixed effects logistic regression with a random intercept for race.

Table 7: Mixed effects model fixed effects

Term	Odds Ratio	p-value
(Intercept)	0.089	<0.001
Age_centered	1.015	<0.001
I(Age_centeredˆ2)	1.000	0.190
GenderM	0.972	0.078
Year2023	1.176	<0.001
Year2024	1.099	<0.001

Table 8: Fixed effects model vs Mixed effects model: Chi-square test

Model	Parameters	Log-likelihood	LR statistic	Pr(>Chi)
Fixed effects (Model 3)	6	-63260.3	NA	NA
Mixed effects (race intercept)	7	-56690.5	13139.6	<0.001

The mixed effects model reveals substantial variation in qualification rates across races (random intercept variance = 0.92), confirming that course selection is an important factor beyond demographics alone. Notably, the quadratic age term loses significance once race-level variation is accounted for, suggesting that the curved age effect observed in earlier models may partially reflect differences in which age groups tend to enter faster races rather than a true nonlinear biological effect.

Interaction Effects

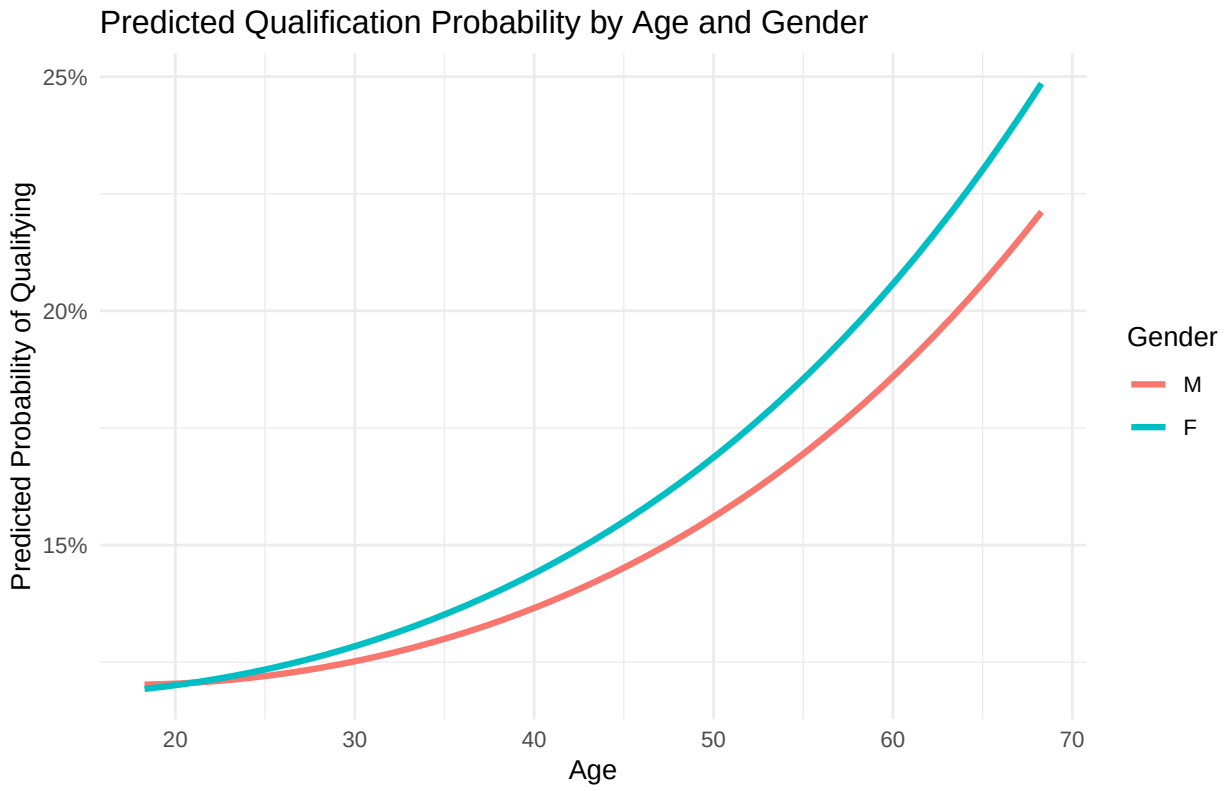
We test whether the effect of age on qualification differs between men and women by adding an age × gender interaction term. The visualization below shows predicted qualification probability across the age range for each gender, holding year constant at 2023.

Table 9: Model 4 coefficients with interaction term

Term	Odds Ratio	p-value
(Intercept)	0.129	<0.001
Age_centered	1.015	<0.001
GenderM	0.946	<0.001
I(Age_centeredˆ2)	1.000	<0.001
Year2023	1.273	<0.001
Year2024	1.271	<0.001
Age_centered:GenderM	0.997	0.005

Table 10: Model 3 vs Model 4: Interaction test

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
159991	126520.6	NA	NA	NA
159990	126512.6	1	8	0.005



Model Evaluation

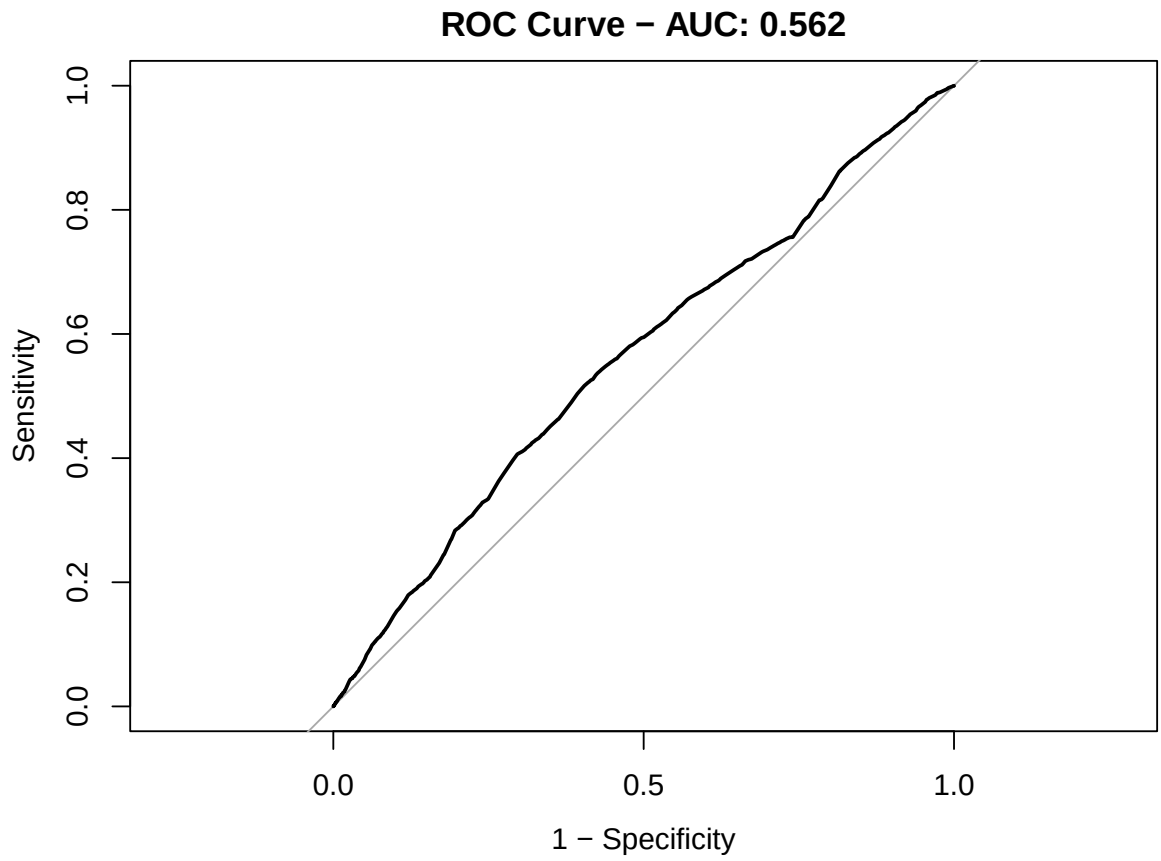
The final model is evaluated on the held-out test set using a classification threshold of 0.14, chosen to reflect the true qualification base rate of 13.6%.

Table 11: Confusion matrix on held-out test set

Predicted	Actual 0	Actual 1
0	23384	3095
1	11211	2310

Table 12: Model evaluation metrics on held-out test set

Accuracy	Precision	Recall	F1
0.642	0.171	0.427	0.244



The held-out test set AUC is **0.562**.

Conclusion

This analysis examined Boston Marathon qualification trends across 1,071,822 runners from 2022 to 2024. Key findings include:

- Qualification rates increased from 11.7% to 14.2% over the study period, accompanied by a 7-minute drop in median finish times.
- The number of runners in the dataset nearly doubled in 2023 because more races were included that year (260 vs. 111 in 2022), not because turnout per race increased.
- Older runners qualify at higher rates, reflecting more lenient age-graded standards, a trend confirmed and quantified by the logistic regression models.
- The age × gender interaction was statistically significant (p=0.005) but small in magnitude (odds ratio 0.997 per year of age), meaning the age effect is slightly weaker for men than for women.
- Year is a significant predictor, with 2023 and 2024 showing higher qualification odds than 2022.
- The logistic regression achieves an AUC of 0.56 using only demographic and temporal features. While modest, this is expected. Without finish time, which cannot be included without introducing target leakage, demographic features alone have limited predictive power. The primary value of this analysis lies in the trend analysis and structural insights rather than individual prediction.